Renko Bar Strategy: SHORT Book, Layer 1 and Layer 2

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Summary

A short-only trend-pullback strategy on 80-tick Renko bricks for MNQ (Micro Nasdaq-100 futures) earned +2.70 points per trade at Layer 1 and +4.30 points per trade at Layer 2 over 12 weeks of tick data (in-sample, after a 0.5-point cost per trade).

The edge comes from one combination: a two-red, one-green brick pullback taken only when three brick EMAs are stacked downward. Without the EMA filter the same pattern lost about 1 point per trade. I designed the research in Python, then built and stress-tested the live strategy in NinjaTrader 8 (C#) on simulation accounts.

Strategy design

Every decision is made at a brick close, from brick data only; time-based bars are not used.

- **Bricks:** 80-tick Renko (20 points on MNQ), with the platform's standard rule that a reversal brick needs twice the brick size.
- **Setup (Filter 1 = `110`):** the last three bricks are red, red, green. In the short book a red brick is coded 1, so `110` reads as two down bricks and a small bounce.
- **Trend gate:** EMA 9 < EMA 20 < EMA 30, computed on brick closes. The setup counts only inside an established down-trend.
- **Entry:** sell at market at the setup brick's close. A limit entry was tested and lowered the result.
- **Exits:** stop 1 brick above the setup close, target 3 bricks below it (1:3 risk to reward). Both are placed as a server-side bracket the moment the entry fills.
- **Size:** 1 contract in the research. The live code supports an optional size rule that reads the day's real results.

Layer 2: filtering on the setup's own track record

Layer 2 trades a setup only if the previous setup lost (Filter 2 = `0`). It cut trades from 514 to 305 and raised the result from +2.70 to +4.30 points per trade.

To make this work, a virtual-trade tracker scores every setup, traded or not, as 1 (target reached), 0 (stop reached) or N (a limit order that never filled). These results form an outcome string, and Filter 2 matches patterns against its tail.

Layer 1 trades every setup. Layer 2 skips the setup right after a virtual win, the point where the next attempt has historically been weakest.

Research results

The EMA gate is what turns the pattern from a loser into a winner, and the edge held in both halves of the sample. Data: MNQ tick data, 12 weeks (July 6 – October 15, 2026), 1 contract, 0.5 point cost per trade, outcomes resolved at brick closes.

Version	Trades	Win rate	Points per trade	Weeks 1–6	Weeks 7–12
Pattern only, no EMA gate	938	26.0%	−1.00	−0.96	−1.11
Layer 1: pattern + EMA gate	514	30.5%	**+2.70**	+2.35	+3.53
Layer 2: Layer 1 + skip after a win	305	32.5%	**+4.30**	+4.06	+4.98
Layer 2 with a 2-brick target	276	42.4%	+3.73	+2.94	+6.08

With a 1:3 payoff the break-even win rate is about 26%, so a 30–33% win rate leaves a real margin. The cost is long losing streaks: runs of 18–25 losses in a row appeared in the sample, so position size has to survive them.

Implementation and testing in NinjaTrader 8

The live strategy is one C# NinjaScript file per book, configured for Layer 1 or 2 by settings alone. Most of the work went into making it fail safely, not into the signal.

- **Same logic live and in research:** a 1,000-brick back-fill at start-up replays history through the exact live code path, so the filters start warm and match the Python research.
- **Virtual-trade tracker:** a win requires a trade at least 1 tick past the target, matching how a real limit target fills.

- **Fast-market handling:** skips an entry already at or through the stop; cancels the rest of a partly filled entry once the stop trades; if a stop is rejected because price has passed it, closes the position and keeps running, with a daily limit that turns the strategy off.
- **Position safety:** a naked-position guard, a wrong-direction watchdog, and closes sized from the strategy's own fill count, because the platform's position updates can lag fills by 1–2 seconds.
- **Start-up and connection safety:** no real order from a stale brick replayed at start-up; every connection change is logged.

Sim testing at 300-contract size on deliberately small bricks surfaced each of these failure modes, and each was fixed and re-tested from the logs. Every test ended flat with no leftover orders.

Limitations and lessons

The results are in-sample: the rules were chosen on the same 12 weeks they are measured on, so forward testing is the next step.

- **Fills:** the simulation assumes stops fill at the stop price; live fast markets fill some stops worse.
- **One market regime:** 12 weeks is a single sample. The split-half check is encouraging but is not out-of-sample proof.
- **Sizing:** increasing size after losses did not improve results in research. A weekly rule based on the previous week's result looked better and stays a manual decision.
- **Lesson:** most outcome-string filters did not add value because trade outcomes are close to independent. The EMA trend gate, an independent piece of information, is what created the edge.